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Hands-on Workshop: USS After Dark: Shell Secrets for Those Who Dare

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What is z/OS USS?

… the brand new thing on Mainframe (introduced ~30y ago)

DID YOU KNOW?



- Since 1993 from OpenEdition MVS -> OS/390 UNIX -> z/OS UNIX System Services
- Allow standard, **open-system applications** (~Java, Node.js, Python, Web Servers) **to run natively on the mainframe** without rewriting them for traditional z/OS envs.
- **POSIX** and not **Linux**. Includes extension for managing EBCDIC and tagging
- **Fully integrated** into z/OS BCP kernel = NOT AN EMULATOR
- **Best of Both Worlds:** UNIX workloads get mainframe superpowers — seamless access to Db2, CICS, and MQ and TSO. Protected by SAF, and prioritized by Workload Manager (WLM). A UNIX process on z/OS is **essentially a z/OS Address Space.**

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Deploying Kafka on z/OS USS

Overcoming legacy constraints with Enhanced ASCII, proper tagging, and modern tooling.

~45min

Environment setup, download, and classification.

1

Enhanced ASCII

Configure USS terminal to handle open source ASCII files

10 Minutes

2

Extraction

⇓ Kafka archive directly to the MF extract without corruption or tripping

10 Minutes

3

Art of File Tagging

Recursively apply tags to the Kafka distribution so Java JVM and USS utilities can read them

15 Minutes

Script modification and live execution.

4

Broker Launch

KRaft formatting and live messaging test.

10 Minutes



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Two Worlds Collide (The Culture Shock)

The Traditional z/OS Developer		The Distributed UNIX Admin
"Where are my records?" Files are byte-streams, not structured datasets (no LRECL/BLKSIZE). No PDS members - folder structure like on your windows desktop	Files & Storage	**"Why is my disk full?"** Filesystems aren't endless disk partitions. They are zFS files (VSAM Linear Data Sets) bound by mainframe allocation rules and extents.
"Who is UID 0?" Moving away from RACF group profiles to Read/Write/Execute (rwx) bits and octal permissions for "owner", "group" and "others"	Security & Power	**“What do you mean root isn't God?"** UID 0 gives UNIX power, but the mainframe security product (RACF/ACF2/TSS) is the ultimate authority. You can be root and still get an access violation.
"Why does everything care about upper and lower case?" File.txt and file.txt are completely different things.	Data Encoding	**"Why does my text look like alien symbols?"** The default encoding is EBCDIC, not ASCII or UTF-8. iconv and file tagging become a way of life.
"How do I submit this?" Shell scripts replace JCL, and standard out/error replace the spool (SDSF).	Execution	**"What is an Address Space?"** UNIX processes are actually backed by z/OS tasks and controlled by WLM, making process management much heavier than standard Linux.

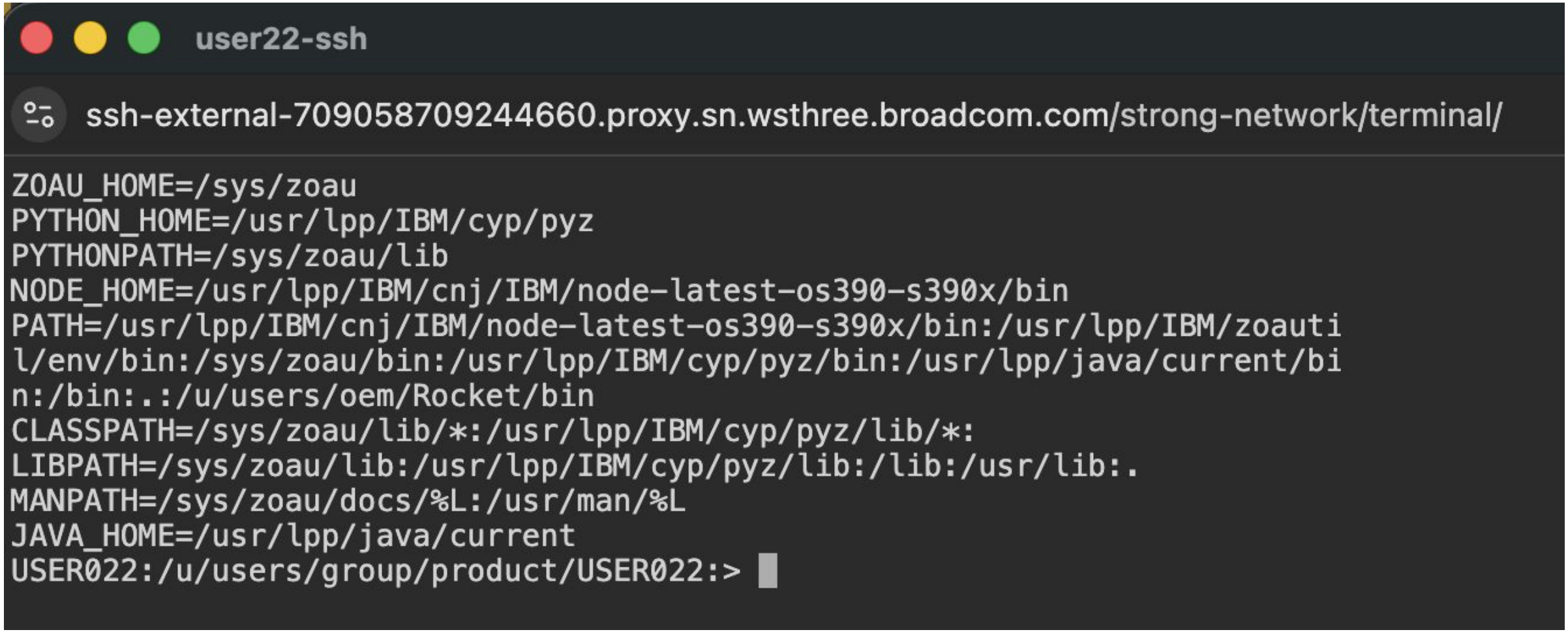
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Workshop Time



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Opening the Shell Environment



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1 Enhanced ASCII

Setting up the shell for auto-conversion

>sh

- z/OS has a default Shell which is Posix Compliant and is not familiar to those who have used linux

>bash

- The default shell on Linux for many years.
- IBM does not distribute bash

>zsh

- Z Shell is the newcomer and is available on z/OS 3.1

Check out https://zopen.community/ Built by IBMers and part of the Open Mainframe Project for bash and other utilities

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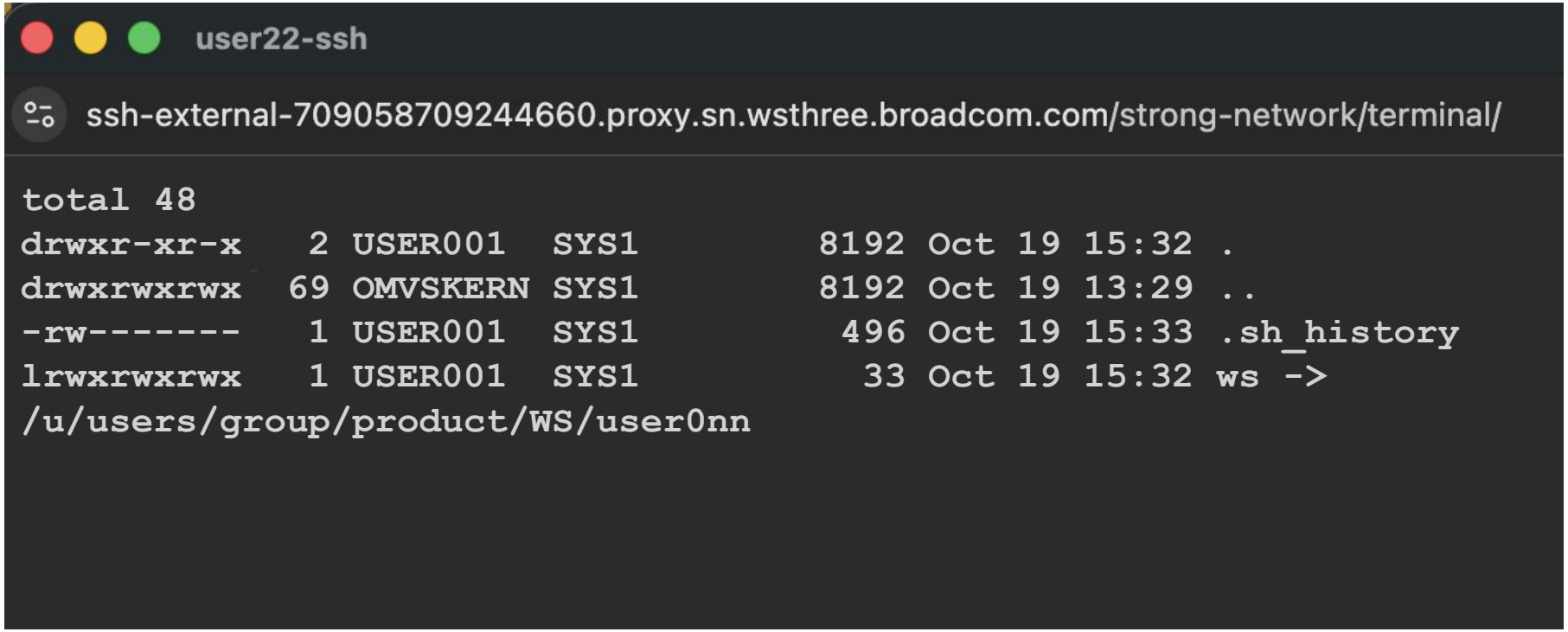
1 Enhanced ASCII

Setting up the shell for auto-conversion

- We will be extracting Kafka from a binary downloaded from the Apache

- In your user directory there is a sub-directory called **ws** and this is where we'll be working.

```
>ls -la
```



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1 Enhanced ASCII

Setting up the shell for auto-conversion

```
>df -k | grep WS
```



This command displays disk space usage in kilobytes, specifically filtering for file systems, directories, or mount points that contain the uppercase letters "WS".

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Enhanced ASCII

Accessing TSO

Let’s leverage TSO from USS

```
>tsocmd "listc ent('zdnt.ws.zfs') all"
```

```
listc ent('zdnt.ws.zfs') all
CLUSTER ------- ZDNT.WS.ZFS
     IN-CAT --- ZDNTCAT.USERCAT
     HISTORY
       DATASET-OWNER----IBMUSER
CREATION--------2026.291
       RELEASE----------------2
EXPIRATION------0000.000
     SMSDATA
```


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Extraction

Deploying the Kafka software in USS for deployment

Background

Goal

Tasks

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2 Extraction

Deploying the Kafka software in USS

- The kafka tar file is located in **`/var/kafka.tar`**
- To extract the archive do the following:

```
>cd ws
```

```
>tar xvf /var/kafka.tar
```

A new directory will be created in your directory = ***`kafka_2.13-3.6.1`***

`>cd` into the directory

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2 Extraction

Deploying the Kafka software in USS

The archive contains either binary files as well as configuration and script files.

We can convert them to EBCDIC but a **better option is tagging them as ASCII.**

- **First**, let's edit one of the script files

```
>vi bin/kafka-server-start.sh
```



To exit the vim editor issue `:q` *and confirm*

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Extracting Kafka

By default, z/OS does not have xterm256-color as a profile. To fix it we:

```
export TERM=vt100
```

Let’s try again:

```
vi bin/kafka-server-start.sh
```


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2 Extraction

Deploying the Kafka software in USS



By default, z/OS does not have **xterm256-color** as a profile.

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```


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…hmm...not the output I wished for…

user22-ssh

ssh-external-709058709244660.proxy.sn.wsthree.broadcom.com/strong-network/terminal/

```
^23^21^G^62^69>^G^62/^73^68^0A^23^20<^69^63^65>^73^65^64^20^74?^20^74^68/
^74^69?>^20^28^41^53^46^29^20^75>^64^65^72^20?>^65^20?^72^20_?^72^65^0A^^
72^65^65_^65>^74^73^F^20^20^53^65^65^20^74^68^65^20+|^54^49^43^45^20^66^A
^23^20^74^68^69^73^20^77?^72,^20^66?^72^20/^64^64^69^74^69?>/%^20^69>^667
^68^74^20?^77>^65^72^73^68^69^70^F^0A^23^20^54^68^65^20^41^53^46^20%^69^5
^20^75>^64^65^72^20^74^68^65^20^41^70/^63^68^65^20<^69^63^65>^73^65^2C^2^
63^65>^73^65^22^29^3B^20`?^75^20_/`^20>?^74^20^75^73^65^20^74^68^69^73^2^
65^20^77^69^74^68^0A^23^20^74^68^65^20<^69^63^65>^73^65^F^20^20^59?^75^29
^63^65>^73^65^20/^74^0A^23^0A^23^20^20^20^20^68^74^74^70^3A^G^G^77^77^77+
^53^45^E^V^F^30^0A^23^0A^23^20^55>%^65^73^73^20^72^65^71^75^69^72^65^64^^
65^64^20^74?^20^69>^20^77^72^69^74^69>^67^2C^20^73?^66^74^77/^72^65^0A^2^
68^65^20<^69^63^65>^73^65^20^69^73^20^64^69^73^74^72^69^62^75^74^65^64^2^
20^57^49^54^48|^55^54^20^57^41^52^52^41+^54^49^45^53^20|^52^20^43|+^44^46
5^72^20^65^78^70^72^65^73^73^20?^72^20^69_^70%^69^65^64^F^0A^23^20^53^650
^73^70^65^63^69^66^69^63^20%/>^67^75/^67^65^20^67?^76^65^72>^69>^67^20^79
?>^73^20^75>^64^65^72^20^74^68^65^20<^69^63^65>^73^65^F^0A^0A^69^66^20$^^
20^22^55^53^41^47^45^3A^20^24^30^20$^E^64/^65_?>)^20^73^65^72^76^65^72^F2
0^70^72?^70^65^72^74`^U^76/%^75^65)^2A^22^0A^09^65^78^69^74^20^31^0A^66^9
^0A^0A^69^66^20$^20^22^78^24.^41^46.^41^<|^47^34^4A^|&^54^53^22^20^U^20^^
~
~
"kafka_2.13-3.6.1/bin/kafka-server-start.sh" [Incomplete last line]  1 ls
```



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3 The Art of File Tagging
Making Kafka Runnable on z/OS

- The tarball from Apache has a familiar format.

```
|____kafka_2.13-3.6.1
| |____LICENSE
| |____NOTICE
| |____bin
| |____config
| |____licenses
| |____libs
| |____site-docs
```

Files like `.sh`, `.properties`, and no ending are likely text files `.jar`, `.class` and others are binary.

- We'll convert, update and set the status of the files.

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3 The Art of File Tagging
Making Kafka Runnable on z/OS

- Let's **first convert a shell script manually**. We've already seen that in its current form (UTF-8) and **without tagging its not useable**.

```
>cd ~/ws/kafka_2.13-3.6.1/bin
```

- Let's pick **kafka-server-start.sh**
- Tagging the file as iso8859-1 (ASCII equivalent)

```
>chtag -t -c iso8859-1 kafka-server-start.sh
```

- This tells z/OS to treat the content as ASCII.
- List the file:

```
>cat kafka-server-start.sh
```


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3 The Art of File Tagging
Making Kafka Runnable on z/OS

The file is now readable. 

- We need to now make an update to the file to use our **`bash`** shell

Execute this command:

```
>sed '1s|/bin/bash|/u/users/oem/Rocket/bin/bash|'
kafka-server-start.sh > kafka-server-start.sh.tmp
```

Verify the new file `kafka-server-start.sh.tmp` has the new shebang:

```
>head -1 kafka-server-start.sh.tmp
#!/u/users/oem/Rocket/bin/bash
```


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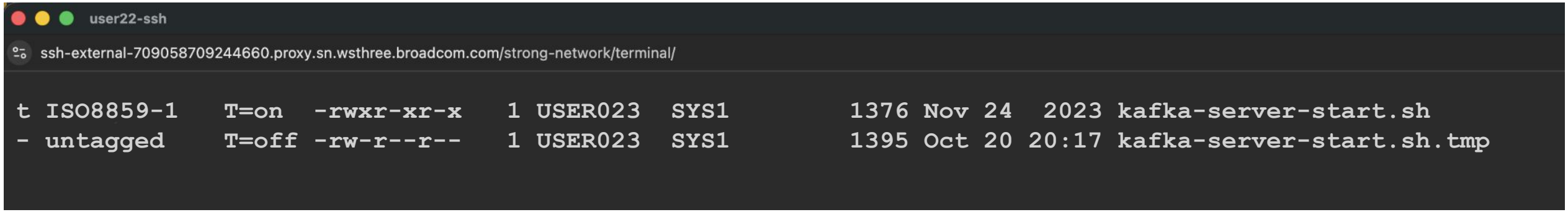
3 The Art of File Tagging
Making Kafka Runnable on z/OS

What we've learned:

We've used `sed` to scan a file and replace a known string with a new one to match our environment.

Execute this command:

```
>ls -laT | grep kafka-server-start
```



Note: the new file is untagged and is in EBCDIC and NOT iso8859-1

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3 The Art of File Tagging
Making Kafka Runnable on z/OS

These are the changes that we need to make although we need to **apply them to the whole tree.**

- We will now apply what we learned **in a single shell script to update the Kafka Installation**.

Execute this file:

```
>/var/tagFiles.sh
```

This will update all `.sh` files and others by tagging them, updating them and **allowing them to be useable in z/OS USS**.

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4 Broker Launch

KRaft formatting and live messaging test.

Let’s see how we did by starting the server



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